

Design real *hardware* with us.

We're looking for students to help add cellular + SIM connectivity to our air quality sensor PCB.

GUARANTEED INTERNSHIP

The project

Our air sensor needs a gateway to get online.

We want to give it a *SIM card* so it works anywhere.

ESP32-H2 · KICAD · LTE-M MODEM · SIM + ANTENNA

What you'd work on

01 Modem integration

Keep the ESP32-H2 as the sensor controller and connect an LTE-M modem over UART, with enable/reset lines. Check I/O voltages and add level shifting where needed.

02 SIM + antenna

Add a SIM (or eSIM) holder, a cellular antenna and its matching/feed, and ESD protection.

03 Power design

Add a dedicated modem rail (around 3.8 V, with transmit peaks near 1 A) and make sure the USB-C input budget holds up.

04 Layout + review

Place and route the new parts in KiCad, run DRC, and get the board ready for fabrication.

05 Bring-up

Help test the first boards: power, SIM detection, network attach.

Who we're looking for

YOU

- **GPA of 3.7 or higher** **REQUIRED**
Include your transcript when you apply
- **EE, ECE or CompE student**
Or anyone who has designed and built a board
- **Some PCB design experience**
KiCad preferred; Altium or Eagle is fine
- **Comfortable with datasheets**
Pinouts, power requirements, reference designs
- **Bonus**
RF or antenna layout, power electronics, embedded C

WHAT YOU GET

★ **A guaranteed internship**

Work on this project with us and you have an internship

- **Real hardware experience**
A board that gets fabricated and used, not a class project
- **Direct mentorship**
Work alongside our team on design reviews
- **A portfolio piece**
Something concrete to show at every interview after this

Interested?

career@noso.so

INCLUDE IN YOUR EMAIL

01

Resume

Year and major

02

Transcript

GPA of 3.7 or higher

03

A board you've made

Photos or design files



THEN

A quick chat

We'll reply to find a time